

Project Design Phase-II

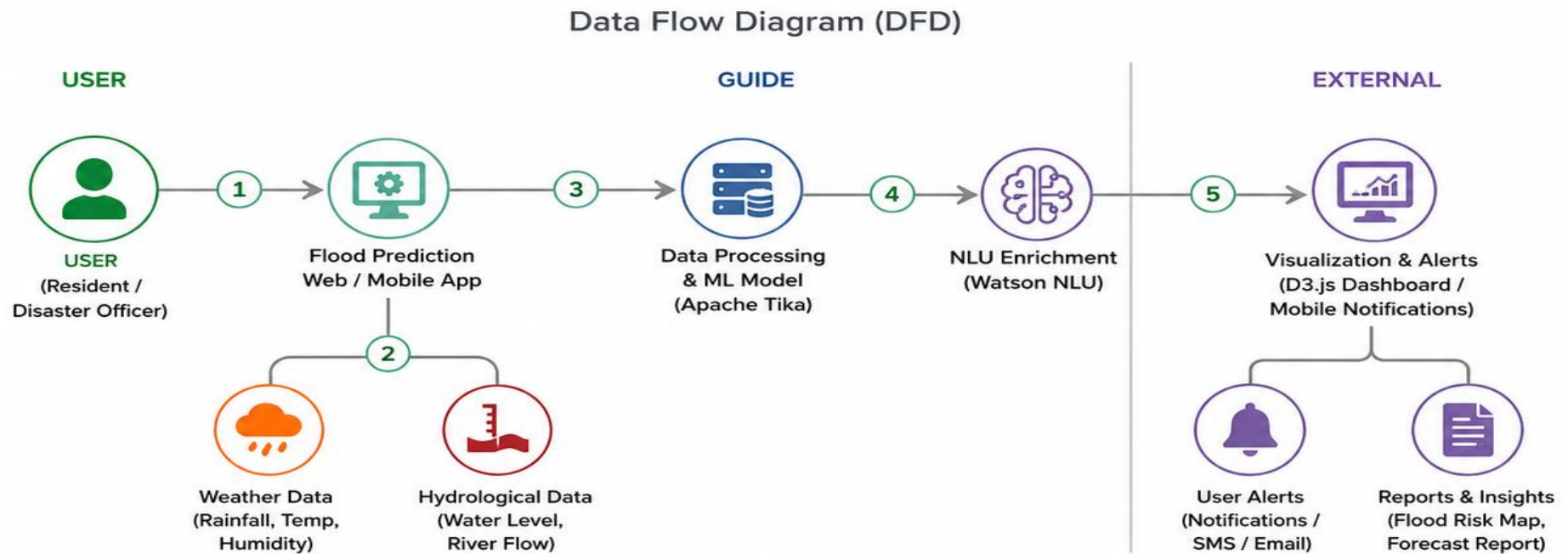
Data Flow Diagram & User Stories

Date	04 July 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	4 Marks

Data Flow Diagrams:

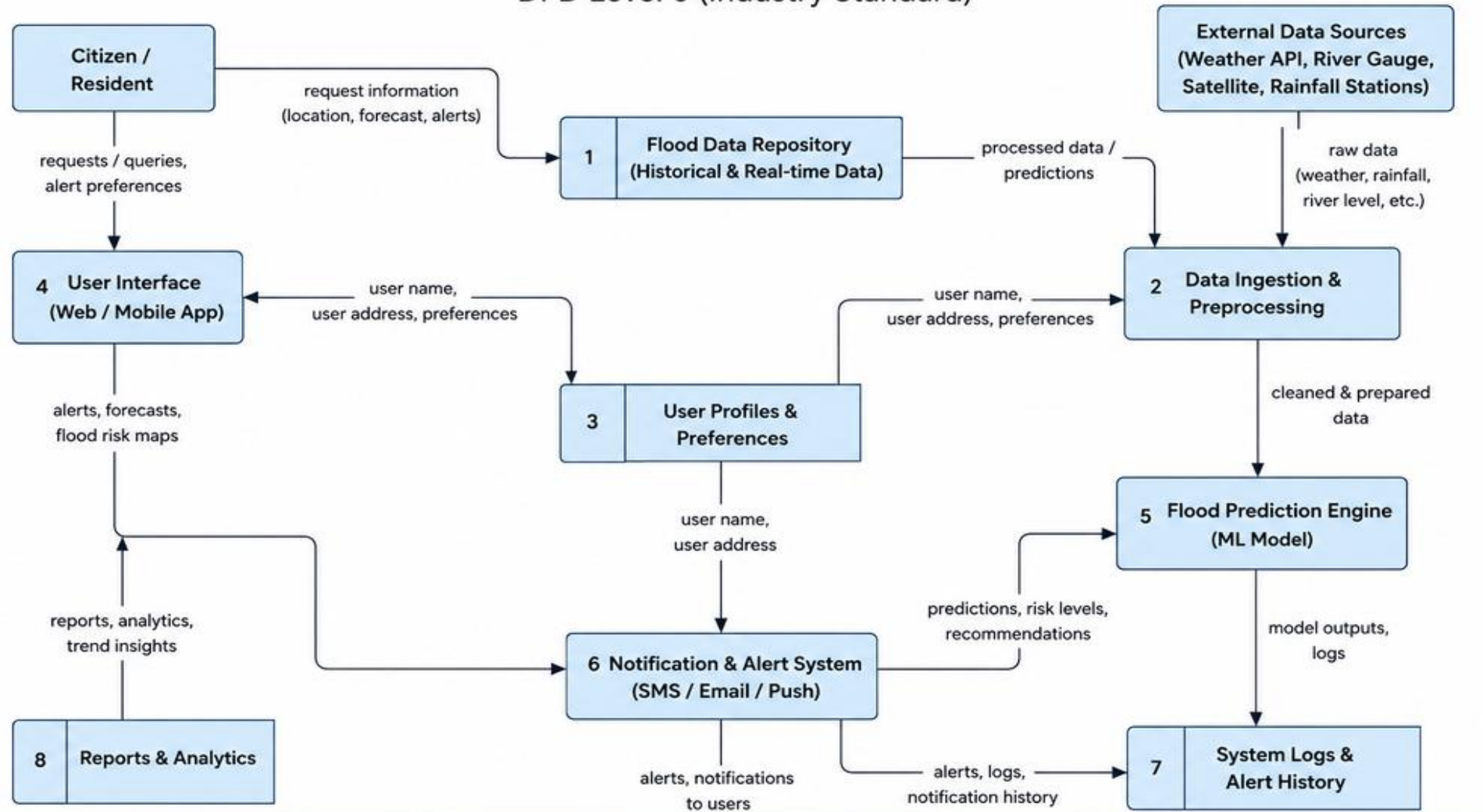
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: [\(Simplified\)](#)



Rising Waters: A Machine Learning Approach to Flood Prediction

DFD Level 0 (Industry Standard)



Use the below template to list all the user stories for the product.

User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Mobile User)	Registration	USN-1	As a user, I can register using my name, email, phone number, password, and location.	User account is successfully created and accessible.	High	Sprint-1
		USN-2	As a user, I receive an email verification after registration.	User receives and verifies email successfully.	High	Sprint-1
	Login	USN-3	As a user, I can log in using my email and password.	User is redirected to the dashboard after successful login.	High	Sprint-1
	Dashboard	USN-4	As a user, I can view my area's current flood risk level and weather conditions.	Dashboard displays real-time flood status and weather information.	High	Sprint-1
	Flood Alerts	USN-5	As a user, I receive instant flood alerts based on my location.	User receives push notifications or SMS before flood events.	High	Sprint-1
	Location Management	USN-6	As a user, I can update my location to receive accurate flood predictions.	System updates flood alerts based on the selected location.	Medium	Sprint-2
	Risk Map	USN-7	As a user, I can view flood-prone areas on an interactive map.	Flood risk map loads correctly with current risk levels.	High	Sprint-2
		Emergency Contacts	USN-8	As a user, I can save emergency contact numbers.	Emergency contacts are stored and accessible during emergencies.	Medium
		Notification Settings	USN-9	As a user, I can enable or disable flood alert notifications.	Notification preferences are saved successfully.	Medium
		Forecast History	USN-10	As a user, I can view previous flood predictions and weather history.	Historical data is displayed accurately.	Low

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Web User)	Login	USN-11	As a user, I can log in through the web portal.	User accesses the web dashboard successfully.	High	Sprint-1
	Dashboard	USN-12	As a user, I can monitor live weather data and flood prediction reports.	Dashboard updates with real-time information.	High	Sprint-1
	Reports	USN-13	As a user, I can download flood prediction reports in PDF format.	Reports download successfully.	Medium	Sprint-2
	Risk Maps	USN-14	As a user, I can view interactive flood-risk maps.	Maps display prediction results accurately.	High	Sprint-2
	Prediction Search	USN-15	As a user, I can search flood predictions by district or city.	Search returns accurate prediction results.	Medium	Sprint-2
Customer Care Executive	User Support	USN-16	As a customer care executive, I can view user complaints.	Complaints are displayed correctly.	High	Sprint-2
	Alert Support	USN-17	As a customer care executive, I can verify flood alerts reported by users.	Alert status can be updated successfully.	Medium	Sprint-2
	Communication	USN-18	As a customer care executive, I can send emergency announcements.	Messages are delivered to affected users.	High	Sprint-3
Administrator	Login	USN-19	As an administrator, I can securely log in to the admin portal.	Admin dashboard opens after successful login.	High	Sprint-1
	User Management	USN-20	As an administrator, I can add, edit, or delete user accounts.	User records are updated successfully.	High	Sprint-1
	Data Management	USN-21	As an administrator, I can upload rainfall, river level, and weather datasets.	Datasets are uploaded and stored successfully.	High	Sprint-2
	Machine Learning Model	USN-22	As an administrator, I can train and update the flood prediction model.	Model retraining completes successfully with updated accuracy.	High	Sprint-3
	Alert Management	USN-23	As an administrator, I can configure flood alert thresholds.	Alert thresholds are saved and applied.	High	Sprint-2
	Report Management	USN-24	As an administrator, I can generate flood analysis reports.	Reports are generated successfully.	Medium	Sprint-3
	System Monitoring	USN-25	As an administrator, I can monitor system health and prediction performance.	Dashboard displays system status and logs.	Medium	Sprint-3
	Audit Logs	USN-26	As an administrator, I can view user activity and system logs.	Audit logs are available for review.	Low	Sprint-3

1. User configures location, preferences, and alert settings in the Flood Prediction App and starts the app.
2. The app collects and selects real-time data (e.g., rainfall, temperature, humidity) and hydrological data (e.g., river water level, flow rate) from various sources to process and load.
3. Apache Tika extracts and parses raw data (text, CSV, JSON, XML, etc.) from the data files.
4. Extracted text/data is passed to Watson NLU for natural language understanding and enrichment (e.g., intent detection, entity extraction, sentiment, location normalization).
5. Enriched data is visualized in the UI using the D3.js library as dashboards, flood risk maps, charts, and alerts.